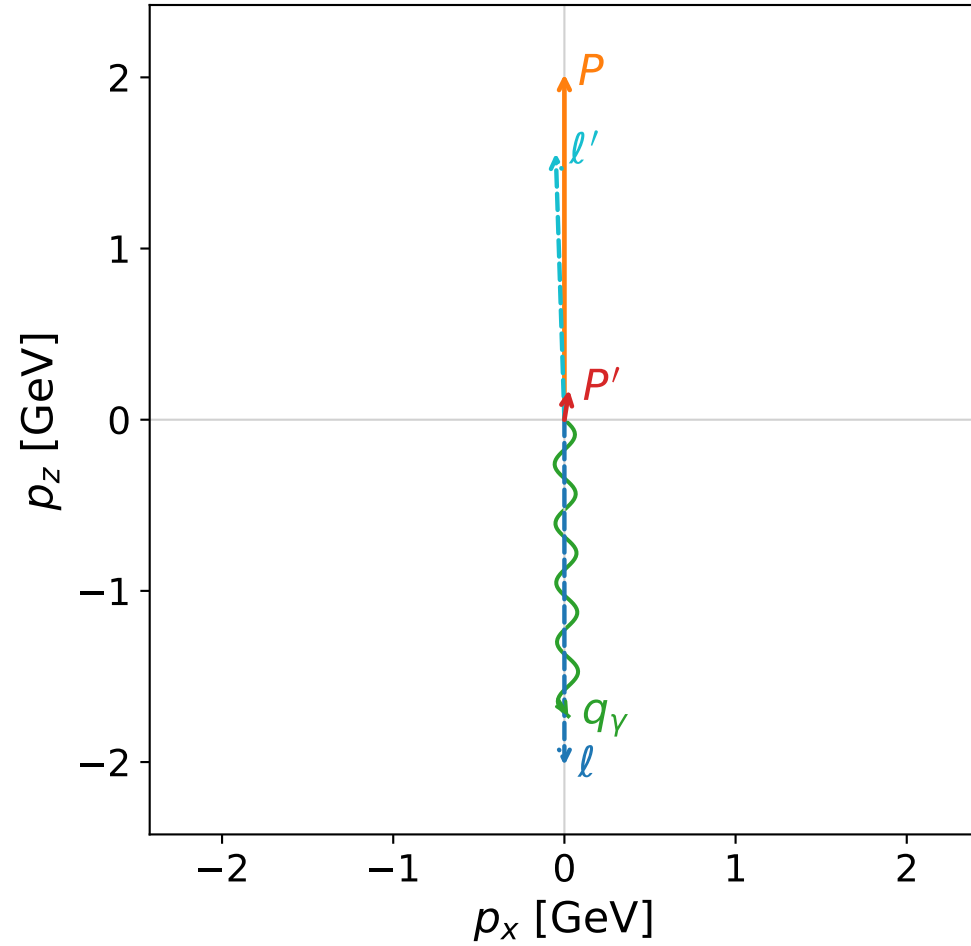
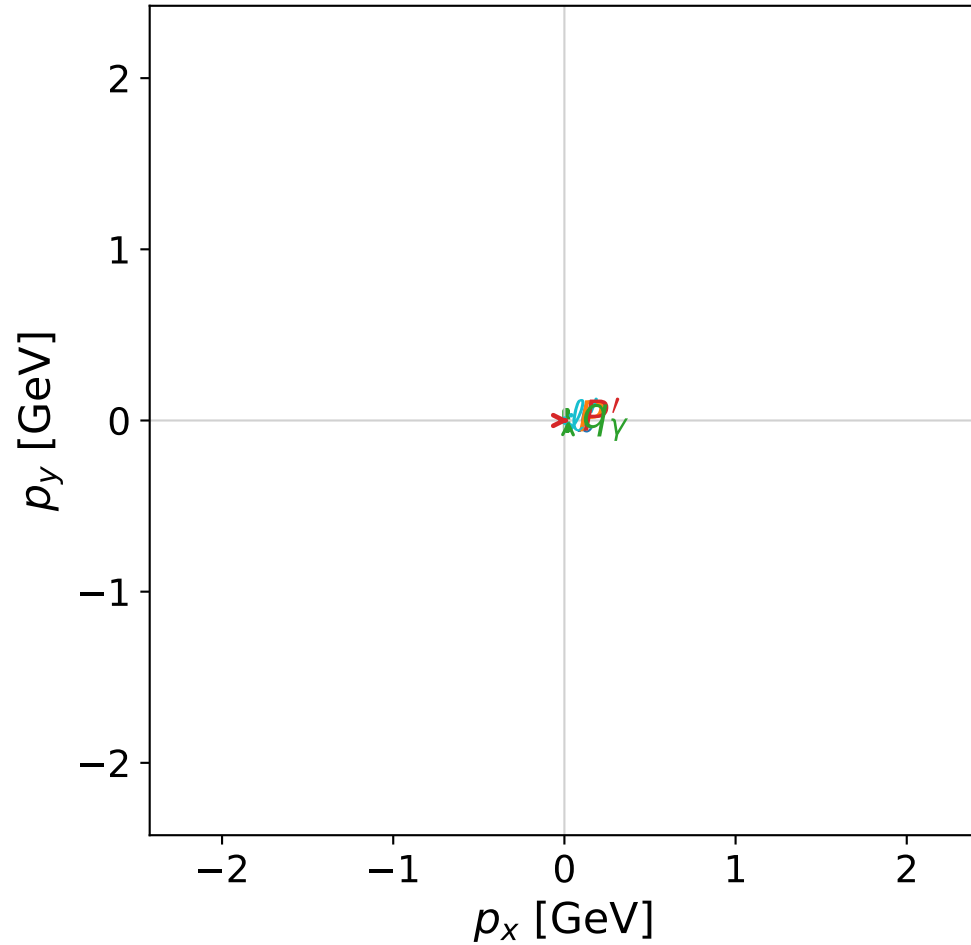


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



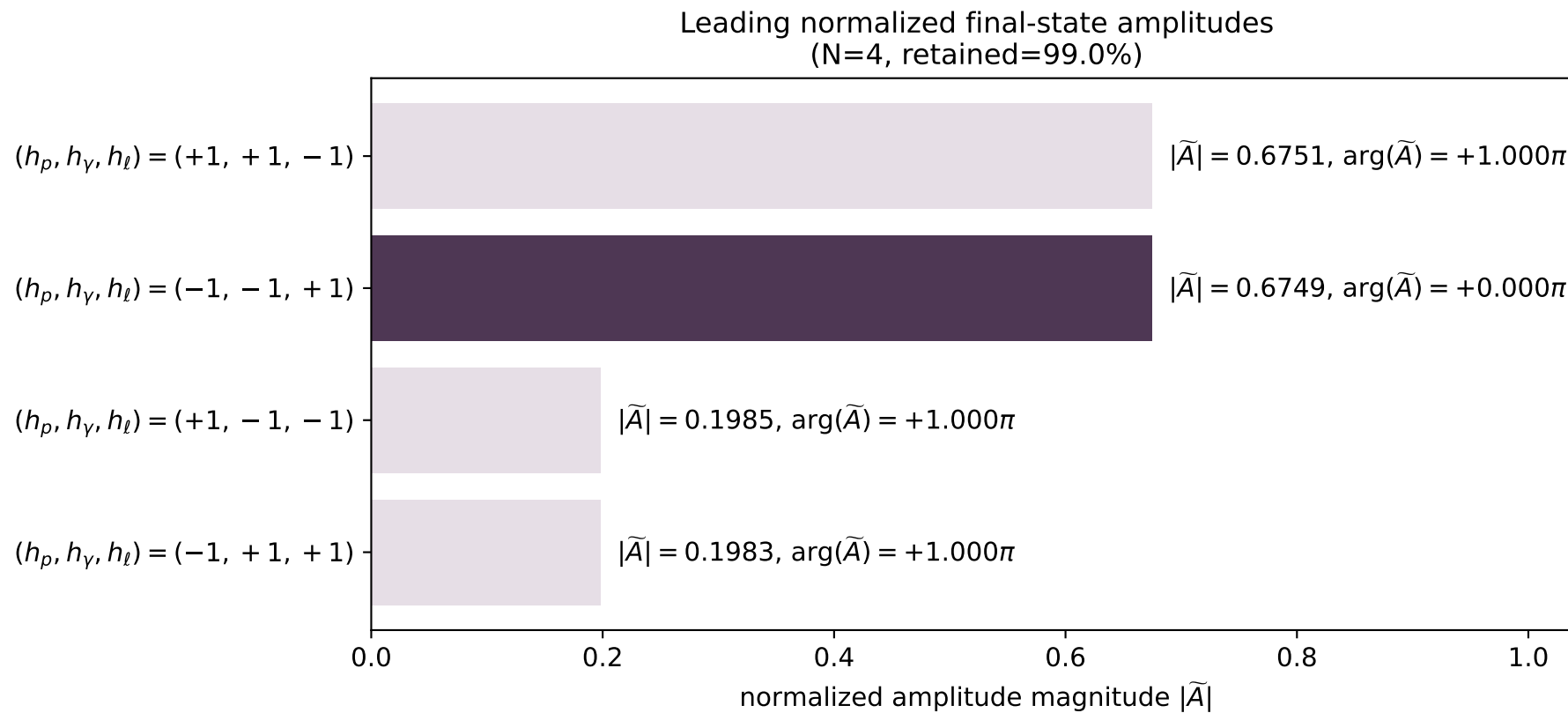
x--y plane



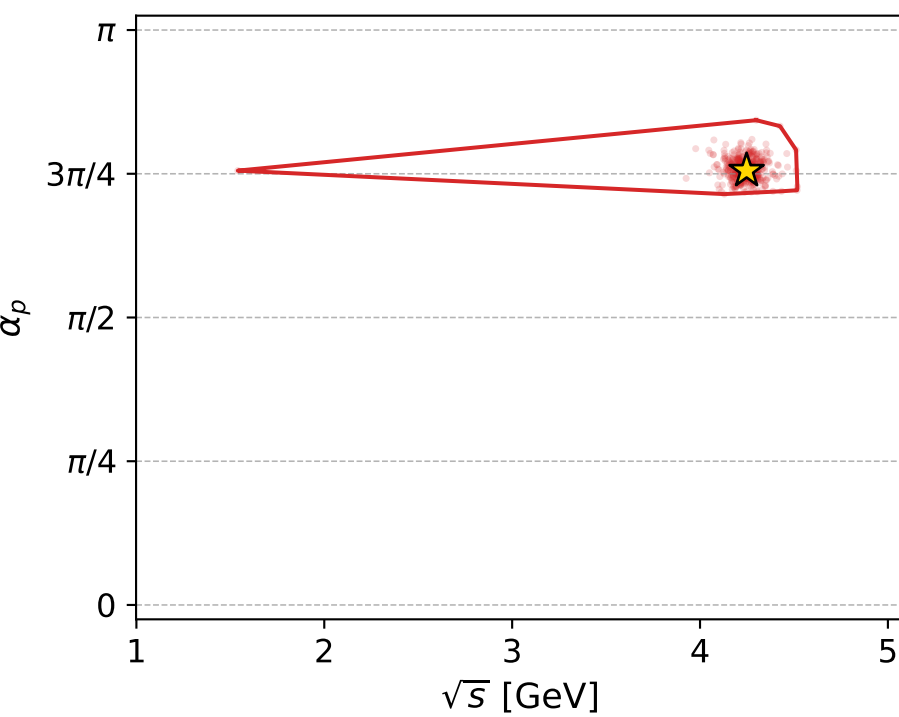
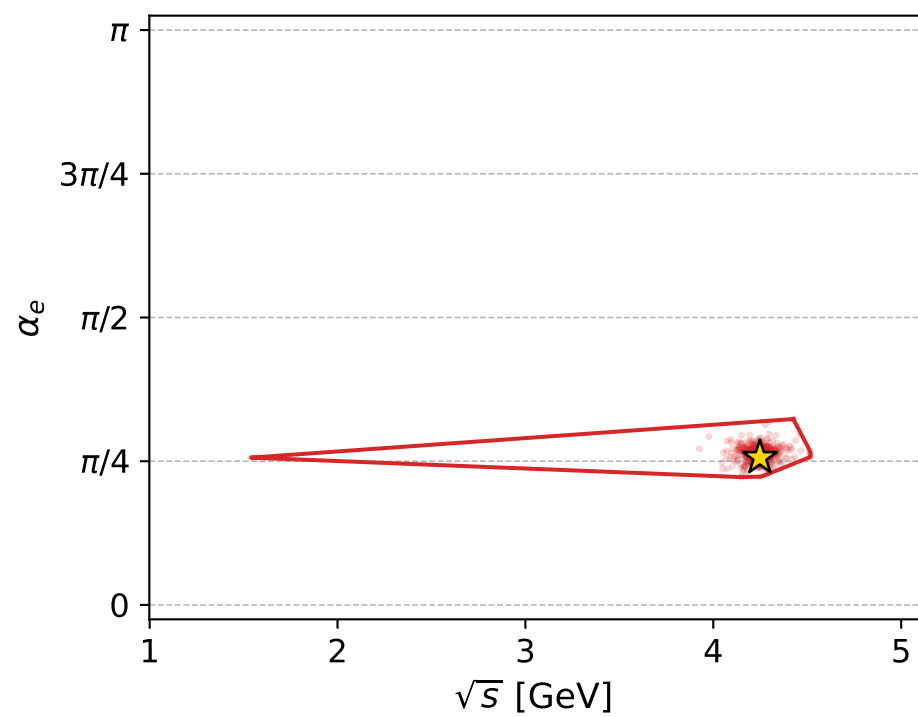
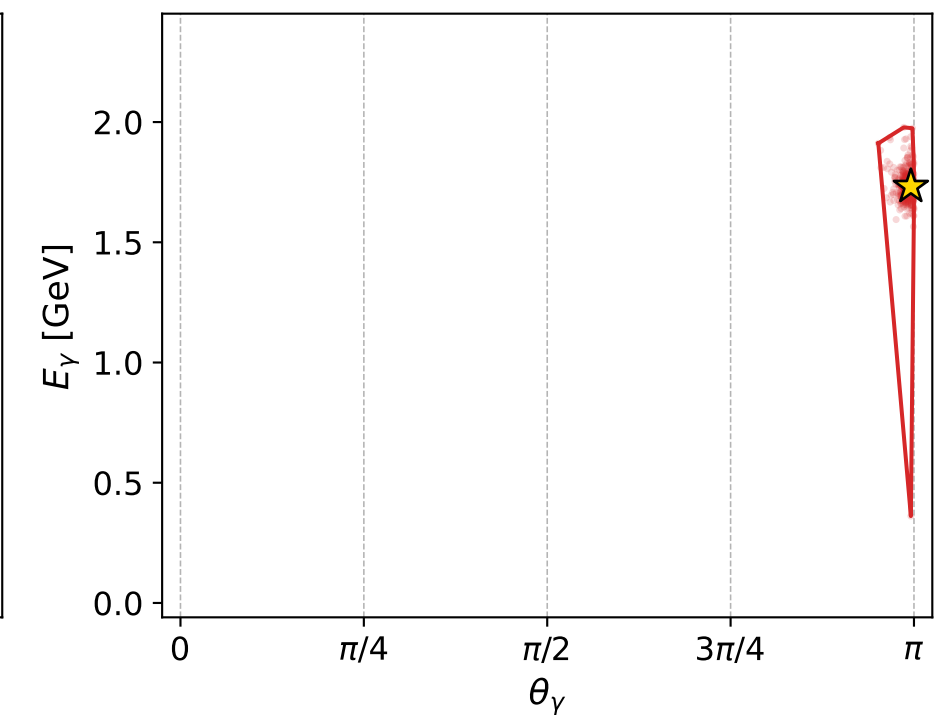
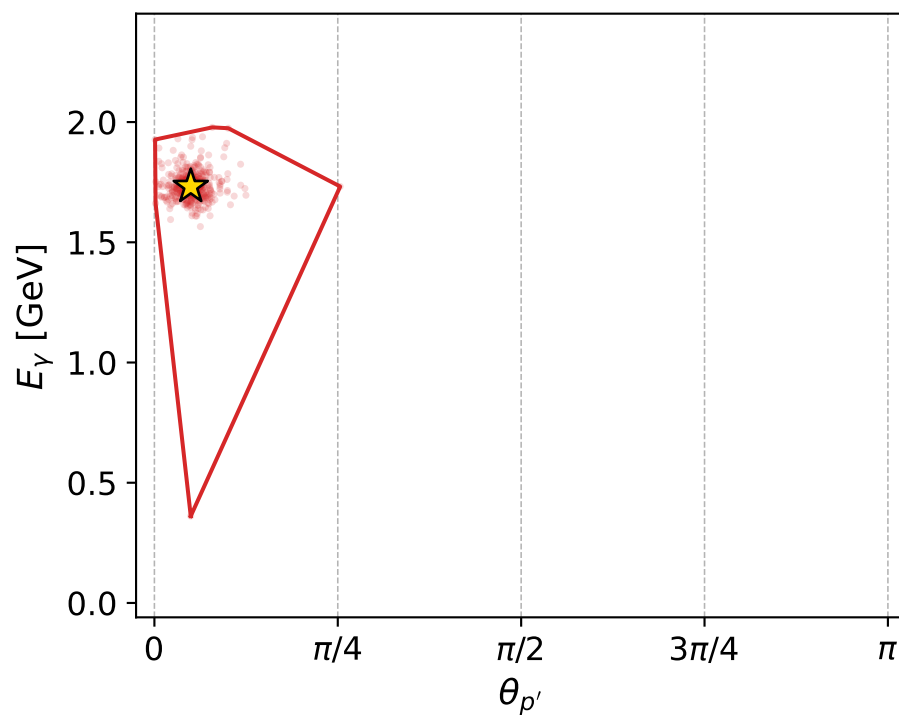
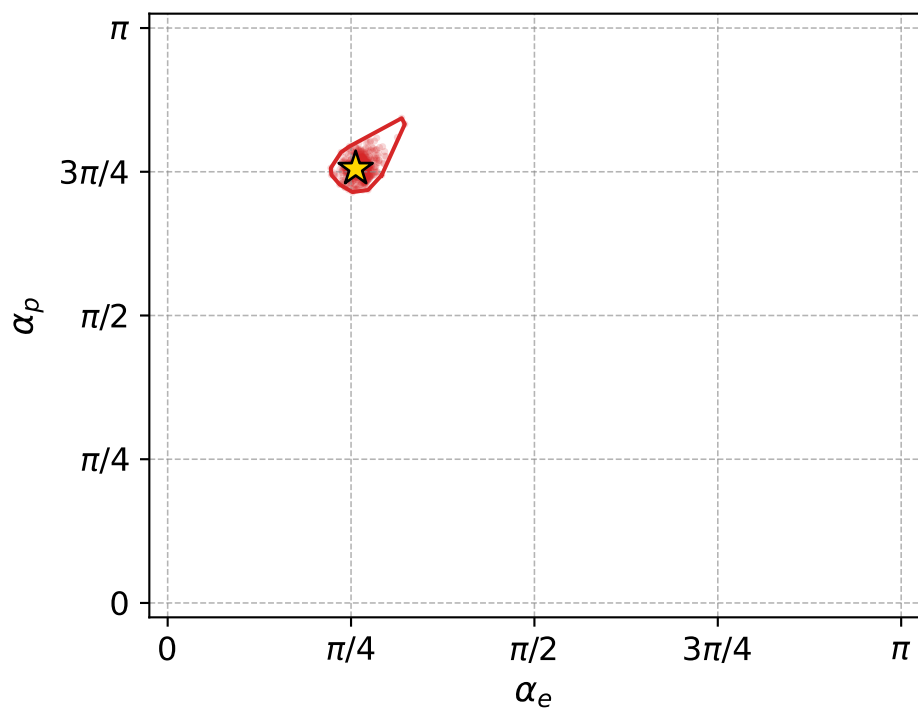
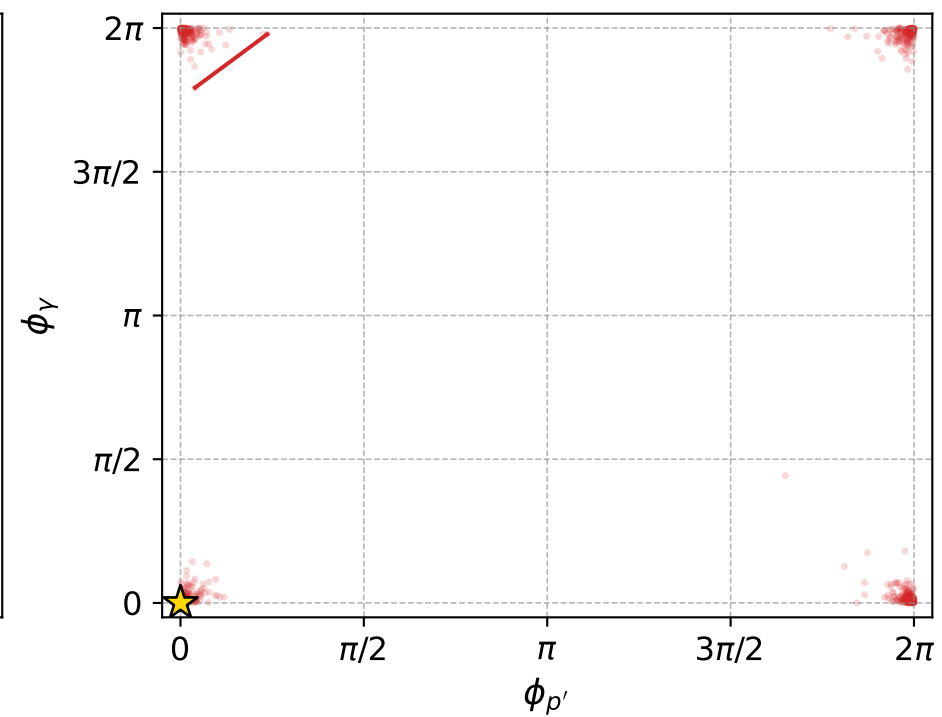
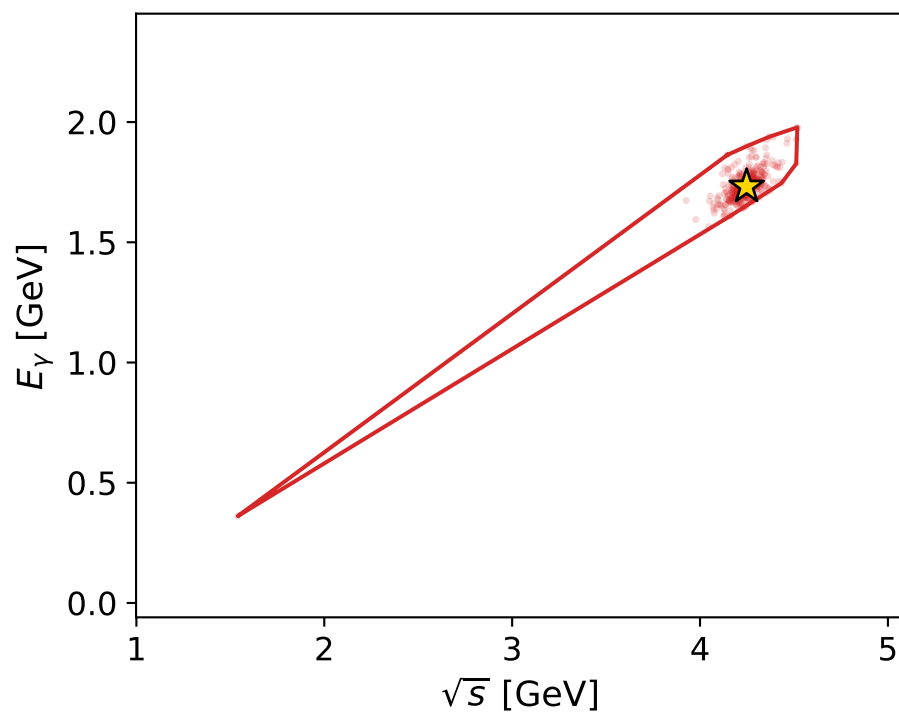
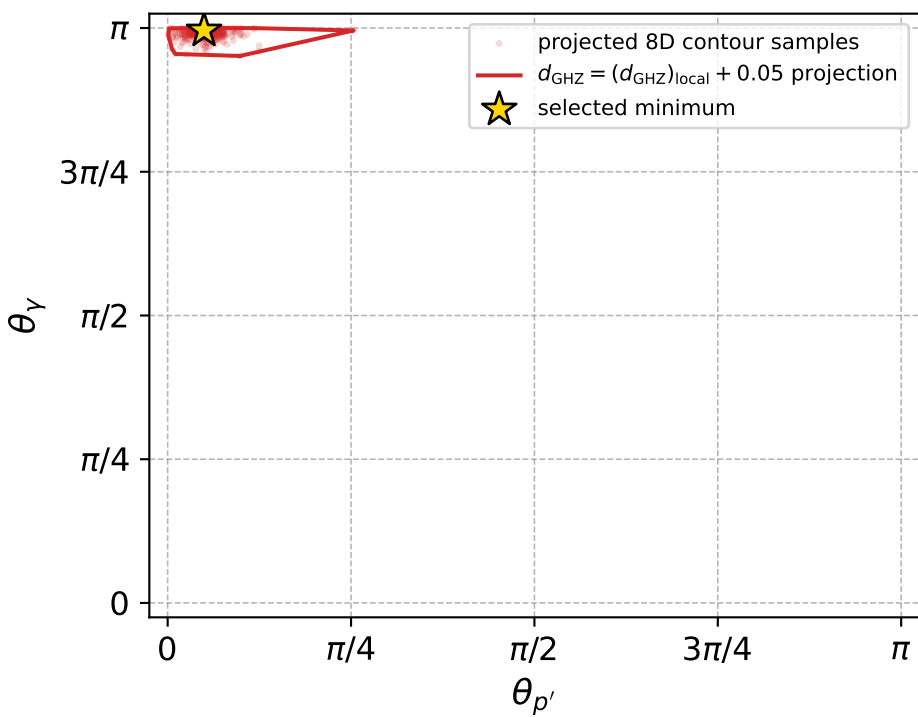
dghz_mixing_angles_polarization_02_local_minimum_0
 $d_{\text{GHZ}}=4.97039\text{e-}07$
 region: polarization_cluster_02_local_minimum_0
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80514431$ rad
 incoming lepton: $0.693007|+\rangle + 0.720931|-\rangle$
 proton mixing angle: $\alpha_p=2.3733154$ rad
 incoming proton: $-0.719109|+\rangle + 0.694897|-\rangle$

$s=18.0448$, $\sqrt{s}=4.24792$, $|\vec{P}|=2.01895$, $|\vec{P}'|=0.177132$
 $\theta_{P'}=0.155688$, $\theta_Y=3.12831$, $\phi_{P'}=0.000235777$, $\phi_Y=2.43077\text{e-}05$
 $E_Y=1.73204$, $\theta_{\ell'}=0.0324071$, $\phi_{\ell'}=3.14173$
 $Q^2=12.5773$, $x_B=0.762773$, $t=-1.7839$, $W^2=4.79145$, $y=0.961237$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0189$
 P $E= 2.2262$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0189$
 ℓ' $E= 1.5613$ $p_x= -0.0505$ $p_y= -0.0000$ $p_z= 1.5569$
 P' $E= 0.9546$ $p_x= 0.0275$ $p_y= 0.0000$ $p_z= 0.1750$
 q_Y $E= 1.7320$ $p_x= 0.0230$ $p_y= 0.0000$ $p_z= -1.7319$



muon: polarization cluster P2, local minimum 0; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.9703864\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000497$
 above global minimum = 0
 8D contour samples = 904
 configured 8D directions = 1536

selected phase-space point:

$\text{sqrt_s} = 4.247919$
 $\text{theta_p_out} = 0.1556878$
 $\text{theta_gamma_out} = 3.128309$
 $\text{q0ut} = 1.732043$
 $\text{phi_p_out} = 0.0002357767$
 $\text{phi_gamma_out} = 2.430766\text{e-}05$
 $\text{alpha_e} = 0.8051443$
 $\text{alpha_p} = 2.373315$